

Focus Hub — Project Overview

A calm, neurodivergent-friendly productivity engine.

Author: Jay Woo · **Live:** whattodo-sable.vercel.app · **Stack:** Vanilla JS · Firebase · Vercel · Chrome MV3 · PWA

1. What it is

Focus Hub is a single-page productivity engine that unifies a task bank, a Google-Calendar-style calendar, a focus timer, and a set of executive-function "scaffolds" designed for ADHD-pattern brains — all in one calm, Notion-inspired interface. It runs as a website, installs to your home screen as a Progressive Web App, and ships with a Chrome extension that floats a translucent quick-capture panel over any page.

It is built for the kind of high-achieving student or builder who has more ambition than their working memory can hold at once — but it works just as well as a clean, classic todo list for anyone who turns the scaffolds off.

2. The problem it solves

Most productivity tools optimize for the **storage** of intent: a place to write things down. Focus Hub optimizes for **execution under cognitive load** — the moment-to-moment friction of actually starting, switching, and finishing tasks when your prefrontal cortex is overloaded.

The design brief, in one line:

How might I build a planner that doesn't fight the way my brain actually works?

Every feature is a vote about whose cognitive load the app respects.

3. Core features

Planning & scheduling

- **Task Bank** — a searchable, filterable, drag-to-schedule pool of everything you need to do, with urgency-colored edges, inline subtasks, and per-subtask scheduling.
- **Calendar** — Month / Week / Day views with a custom hour grid (no library), drag-and-drop scheduling, double-click quick-add, pinch / Cmd-scroll zoom, drag-resizable panels, and a fits-in-one-viewport layout.
- **iCal import** — absorbs Google Calendar exports with full recurring-event support (RRULE / EXDATE / RDATE / VTIMEZONE), multi-file import, and per-calendar color palettes.
- **Natural-language Quick Add** — type "MUST submit essay by Friday" and it parses priority, due date, and type automatically.
- **Focus blocks & timer** — reserve time, pick tasks, run a Pomodoro-style focus session.

The five cognitive scaffolds (*all opt-out*)

Scaffold	What it does	The problem it targets
First Step	Every task surfaces its 5-second move ("put left shoe on")	Transition paralysis
Quick Capture	■K from anywhere — 3-word brain dump	Working-memory wipe
Dopamine Gate	Soft-locks "plan" tasks until you log execute work	Productive procrastination
Domain Modes	KTLO vs Active per life domain	Capacity burnout
Time Anchor	Intrusive toast every 25 min during focus	Time blindness

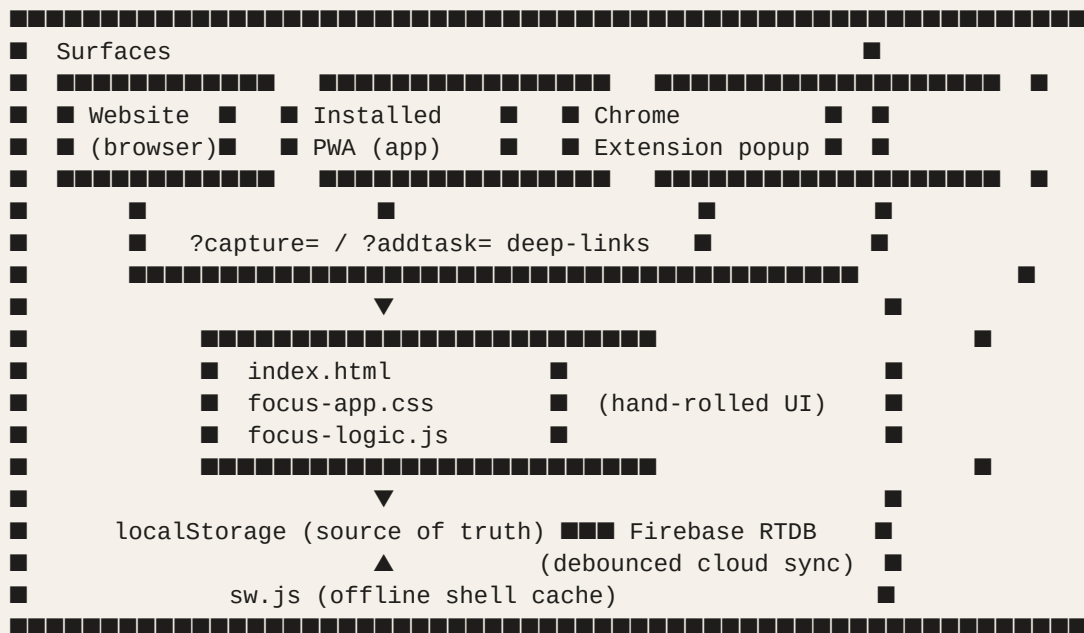
Surfaces

- **Website** — the full app at whattodo-sable.vercel.app
- **Installable PWA** — adds to macOS Dock, Windows Start, iOS / Android home screen; opens in its own window
- **Chrome extension** — a translucent "folded-tab" popup that floats over any page for instant capture

4. Design principles

- 1 **Calm over loud.** Notion-palette pastels, typography never above weight 540, synthetic bold disabled globally. The first screen shows a greeting, the date, and today's tasks — nothing else demanding attention.
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- 4 4
- 5 5

5. Architecture at a glance



- **Persistence:** localStorage is the source of truth; Firebase Realtime Database provides cross-device sync with a 400 ms debounce and a `beforeunload` flush.
- **Service worker:** caches the static shell for instant launches; never caches Firebase or user data.
- **Deep-links:** `?capture=`, `?addtask=`, `?inbox=`, `?shortcuts=` let the extension and PWA shortcuts drive the main app.

6. Technical highlights

- **52+ commits** across a focused build cycle.
- **~6,800 lines of JavaScript, ~4,500 of CSS, ~2,000 of HTML** — all hand-rolled.
- **Zero front-end framework.** Only runtime dependency is Compromise.js for natural-language parsing.
- **Custom calendar hour grid** (~600 lines of pure DOM math) with dynamic per-render pixel sizing so it fits any viewport in one view.
- `requestAnimationFrame` **render coalescing** so rapid completion clicks never thrash the UI.
- **Auto-migrating schema** — every new field gets a safe default for existing user data; never a destructive migration.
- **Full PWA** — manifest with maskable icons, service worker, and a per-platform install flow.
- **Chrome MV3 extension** — translucent liquid-glass popup with `backdrop-filter` blur, dark-mode aware, deep-linking back to the main app.

7. Engineering lessons

- 1 **Local-first is a feature, not a convenience.** I shipped a debounced cloud-sync layer only after being bitten by the exact bug local-first architectures exist to prevent (tasks "disappearing" on refresh when a signed-in user's Firebase write timed out silently).
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8. Roadmap

- **Habit Brain Map** — a force-directed visualization of habit interconnections with AI-proposed merges gated by goal-preservation checks.
- **Extension v2** — live Today's-tasks list in the popup (Firebase fetch), badge count of unprocessed captures, right-click "capture page title + URL."
- **Whisper-based audio transcription** for video ingestion (opt-in, dynamic import).
- **Habit streak heatmaps** colored by theme.

9. Try it

Surface	How
Website	whattodo-sable.vercel.app
Install as app	Open the site → click the "Install" banner (Chrome/Edge) or Share → Add to Home Screen (iOS) / File → Add to Dock (macOS Safari)
Chrome extension	chrome://extensions → Developer mode → Load unpacked → select /extension → press ■ B anywhere

Focus Hub is in active development and private beta. Source, full case study, and a live demo are available on request.